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Fourth Semester- B.Tech. Course work
END SEMESTER EXAMINATION, April-May, 2024

Course code: COMTC13/CAMTC13/CBMTTC13/CDMTTC13/CIMTC13

Course title: Probability and Stochastic Processes

Time: 3 hours

Maximum Marks: 50

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

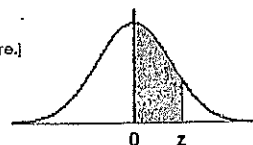
Q. No.	Attempt any two parts from each question	Marks	CO																		
1a	(a) Two cards are drawn at random from 4 cards numbered 1 to 4. Find the expectation and variance of the sum of points on two cards. (b) Suppose that X is a random variable for which $E(X) = 10$ and $Var(X) = 25$. Find the positive values of a and b such that $Y = aX - b$, has mean 0 and variance 1.	3+2	CO1																		
1b	The diameter of an electric cable X is a continuous random variable with cumulative distribution function $F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 3x^2 - 2x^3 & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x \geq 1. \end{cases}$ Answer for: (a) Is X a continuous R.V? Justify? (b) Find $E(X - 1/2)$. (c) Find $P(X \leq \frac{1}{2} < X < \frac{3}{4})$.	5	CO1																		
1c	The first four moments about origin of a distribution are 2, 136, 320, and 40,000. Find the Kurtosis of the distribution curve.	5	CO1																		
2a	A salesman from a travel agency knows from experience that the opportunity to sell a travel package is greater when he makes more contact with potential buyers. He has established that the probability that a customer will buy a package after a visit is constant and equal to 0.01. If the set of visits done by the salesman constitute an independent set of trials, how many potential buyers must he visit in order to guarantee that the probability of selling at least one travel package will equal 0.85?	5	CO2																		
2b	Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50. (a) What can be said about the probability that this week's production will exceed 75 items? (b) If the variance of a week's production is equal to 25, then what can be said about the productivity that will be between 40 and 60?	5	CO2																		
2c	Why do we fit any particular distribution to an underlying data? The following table gives baseball throws for a distance by first year high school girls: <table border="1" style="margin: 10px auto;"> <tr> <th>Distance in feet</th><th>15-25</th><th>25-35</th><th>35-45</th><th>45-55</th><th>55-65</th><th>65-75</th><th>75-85</th><th>85-95</th></tr> <tr> <th>Frequency</th><td>7</td><td>25</td><td>35</td><td>50</td><td>37</td><td>21</td><td>10</td><td>4</td></tr> </table> Fit a normal distribution using method of area and find the theoretical frequencies for the classes of the given frequency distribution:	Distance in feet	15-25	25-35	35-45	45-55	55-65	65-75	75-85	85-95	Frequency	7	25	35	50	37	21	10	4	5	CO2
Distance in feet	15-25	25-35	35-45	45-55	55-65	65-75	75-85	85-95													
Frequency	7	25	35	50	37	21	10	4													
3a	Derive the M.G.F of a normal distribution and hence find out its cumulants of all orders and Kurtosis.	5	CO3																		
3b	State Weak Law of Large Numbers. Let X_i assume that values i and $-i$ with equal probabilities. Show that the Weak Law of Large Numbers cannot be applied to the sequence of independent variables $X_1, X_2, \dots$, i.e., X_i 's.	5	CO3																		
3c	By the method of least square, fit an exponential curve of the form $y = ae^{bx}$ to the following data: <table border="1" style="margin: 10px auto;"> <tr> <th>x</th><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <th>y</th><td>1.6</td><td>4.5</td><td>13.8</td><td>40.2</td><td>125</td><td>300</td></tr> </table>	x	1	2	3	4	5	6	y	1.6	4.5	13.8	40.2	125	300	5	CO3				
x	1	2	3	4	5	6															
y	1.6	4.5	13.8	40.2	125	300															

4a	A random variable X follows the exponential distribution $f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}; x \geq 0, \theta > 0$. A single observation of an exponential random variable X is used to test the null hypothesis $H_0: \theta = 5$ against the alternative hypothesis $H_1: \theta = 10$. H_0 is rejected when the observed value of the random variable is 15 or more. Find the critical region, acceptance region and also find the type I and type II errors.	5	CO4									
4b	An ambulance service claims that it takes on an average 8.9 minutes to reach its destination in emergency calls. To check on this claim, the agency which licenses ambulance services has them timed on 50 emergency calls, getting a mean of 9.3 minutes with a standard deviation of 1.6 minutes. What can they conclude at the level of significance 0.05? Is your conclusion changes if level of significance becomes 0.1?	5	CO4									
4c	In an experiment with immunization of cattle from tuberculosis, the following results were obtained: <table border="1"> <tr> <td></td> <td>Affected</td> <td>Unaffected</td> </tr> <tr> <td>Inoculated</td> <td>12</td> <td>26</td> </tr> <tr> <td>Non-Inoculated</td> <td>16</td> <td>6</td> </tr> </table> Examine the effect of vaccine in controlling susceptibility of tuberculosis at 5% level of significance. [$\chi^2_{0.05,1} = 3.841$, $\chi^2_{0.025,1} = 5.024$, $\chi^2_{0.05,2} = 5.991$, $\chi^2_{0.025,2} = 5.124$]		Affected	Unaffected	Inoculated	12	26	Non-Inoculated	16	6	5	CO4
	Affected	Unaffected										
Inoculated	12	26										
Non-Inoculated	16	6										
5a	The mean weekly sales of candies in candy stores was 146.3 candies per store. After an advertising campaign the mean weekly sales in 22 stores for a typical week increased to 153.7 and showed a standard deviation of 17.2. Was the advertising campaign successful at 1% level of significance?	5	CO5									
5b	You want to rent an unfurnished four-bedroom apartment in Dwarka, ND. The mean monthly rent for a random sample of 60 apartments advertised on "no broker.com" is \$1000. Assume a population standard deviation of \$200. - Find the standard error of the mean? - Construct 95%, 99% confidence intervals and interpret each of the interval?	5	CO5									
5c	(a) A random sample (X_1, X_2, X_3) of size 3 is drawn from a population with mean value μ and variance σ^2 . Which of the T_1 or T_2 are unbiased estimators where $T_1 = 2X_1 + 3X_2 - 4X_3$ and $T_2 = X_1 + X_2 - X_3$? (b) Obtain the MLE of θ in $f(x, \theta) = (1 + \theta)x^\theta, 0 < x < 1$, based on an independent sample of size n .	5	CO5									

Half table for determining the proportion under any chosen section of the normal curve. (The standard distribution table is symmetrical about the centre.)

This table applies to the green area in the graph to the right

Deviation (distance) Z from the mean (average) at 0



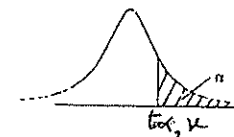
Z	0	1	2	3	4	5	6	7	8	9
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0310	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0754
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2610	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2996	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3829
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4266	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4648	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4755	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4993	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998
3.6	.4998	.4998	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.7	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.8	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999	.4999
3.9	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000	.5000

Figure 1: Standard Normal distribution

The table gives the value of $t_{\alpha, \nu}$ where ν is a degree of freedom such that $Pr(T > t_{\alpha, \nu}) = \alpha$.

Table of the Student's t-distribution

The table gives the values of $t_{\alpha, \nu}$ where $Pr(T > t_{\alpha, \nu}) = \alpha$, with ν degrees of freedom



$\alpha \rightarrow$	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.526	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.659	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Figure 2: Table for t-distribution